Analytical Approaches I

LOGISTIC REGRESSION

Network Analysis in JASP

Introduction

Logistic regression is used when the dependent variable is **qualitative (e.g., Yes/No)**

Recall

A qualitative variable (also known as categorical) is not numerical, and its values fit into categories. It is also divided into two types:

- Nominal: no ordering is possible or implied in the categories (for example *sex at birth*);
- Ordinal: an order is implied in the categories (for example the health status, such as poor/reasonable/good, a Likert scale)

A **binary** variable, also known as dichotomous, is a special case of a qualitative nominal variable when there are **only two categories**.

Research Questions & Methodological Approaches

- Medicine
  - Estimate the risk factors associated with a disease or a harmful condition
  - Predict the risk of developing a disease based on a patient's medical history
  - Determine the most "important" biological factors associated with a specific disease or condition
- Economics
  - Estimate a debtor's likelihood to default on a loan based on creditworthiness
  - Model a customer's propensity to buy a certain product or service given demographic profile
- Psychology
  - Estimate the probability of developing PTSD following trauma
- Political Science
  - A question I examined ad nauseum in graduate school: What predicts voting behavior?

Generalized Linear Models

Two categories: Binary logistic regression is used when the goal is to estimate the relationship between a **binary dependent variable** (= two outcomes), and one or more independent variables (of any type)

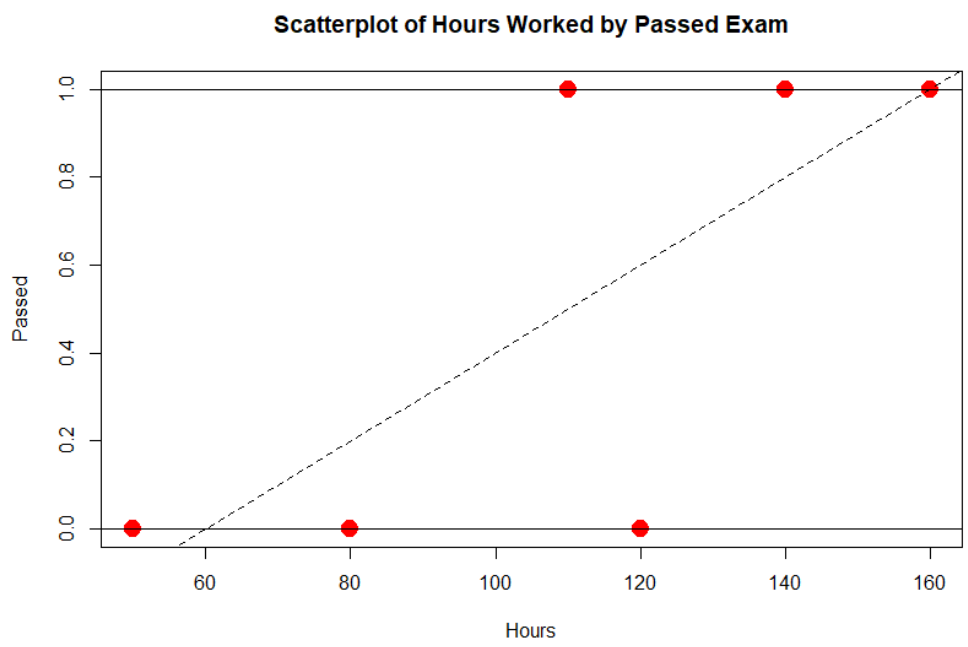
More than two categories: Multinomial logistic regression is used when the goal is to estimate the relationship between a **nominal dependent variable** with three or more *unordered* outcomes, and one or more independent variables (of any type).

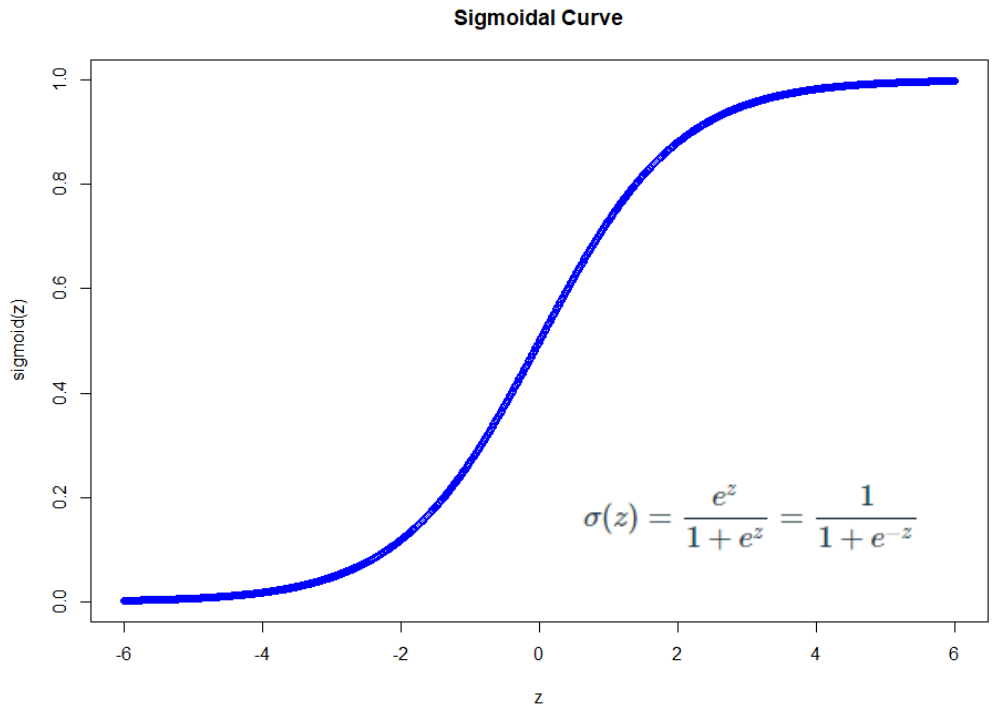
More than two *ordered* categories: Ordinal logistic regression is used when the goal is to estimate the relationship between an **ordinal dependent variable** with three or more *ordered* outcomes, and one or more independent variables (of any type).

Categorical regressions (i.e., of the logistic family) (and Poisson regression) are both part of a broader type of model called **generalized linear models** (abbreviated as **GLM**).

Linear vs. Logistic Regression

The sigmoidal function models the probability that *Y* = 1 conditional on *X* (i.e., values falling along the x-axis); using the **logit function ensures that the Y values will be between 0 and 1 for all values of *X* (i.e., no matter the range of *X*)**





Univariate versus multivariate logistic regression

Univariate binary logistic regression includes **one independent variable**

Multivariate binary logistic regression includes **two ore more independent variables**

- Variables can be quantitative or qualitative

Mechanics of Logistic Regression

A regression where the dependent variable is measured 0/1 (pass/fail; vote/didn't vote, etc.)

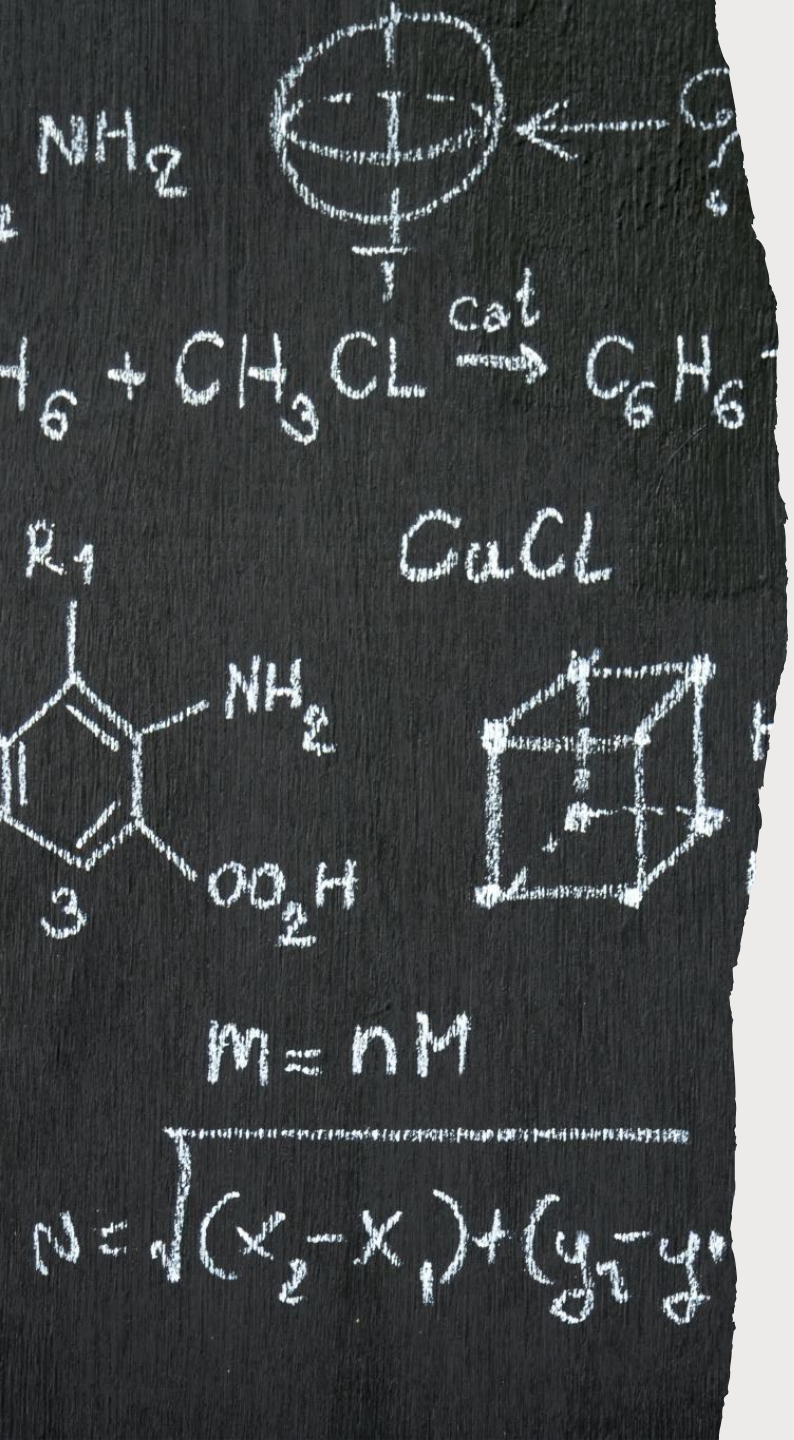
In ordinary regression the model predicts the mean of *Y* for any combination of predictors

Goal of logistic regression: Predict the "true" proportion of success, *p*, at any value of the predictor.

p = Proportion of "Success"

What's the "mean" of a 0/1 indicator variable?

$$\bar{y} = \frac{\sum y_i}{n} = \frac{\#\text{ of } 1's}{\#\text{ of trials}} = \text{Proportion of "success"}$$



Careful

- Binary logistic regression models the probability that Y = 1.
- For example, if *Y* = 1 if the student passed, and 0 otherwise, we are modeling the likelihood that the student passes or not (conditional on the independent variables in the model)
- This likelihood will be framed as a probability

Recall the relation between odds and probability

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_m X_m$$

Odds and logit (or log odds)

Odds = the ratio of the probability of the dependent variable's two outcomes

- Taking the log odds of *Y* creates a linear relationship between *X* and the probability of *Y* (Pampel, 2000)

$$\ln\left[\frac{P(Y=1)}{1-P(Y=1)}\right] = Logit(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_m X_m$$

Equivalent forms of the logistic regression model

Y = Binary response

X = Quantitative predictor

Logit form

$$y = \log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X$$

This is natural log (aka "ln")

Log Odds

Probability form

$$\Pr(Y = 1) = p = \frac{exp^{\beta_0+\beta_1 X_1}}{1 + exp^{\beta_0+\beta_1 X_1}}$$

p = proportion of 1's (yes, success) at any X

Probability

Mechanics of Logistic Regression

- Logit of *Y* equation interpretation: For each one-unit change in the independent variable, the **log odds of Y** increases (decreases) by the beta coefficient
  - **Problem**: nobody knows what log odds are!
- Exponentiate the logit (*Y*) (i.e., the outcome of the logistic regression equation), to convert it back to the odds
  - Interpret the independent variables as affecting the odds (rather than log odds) of the outcome
- Convert odds to probability: Probability values close to one indicate increased likelihood of occurrence
  - Much more intuitive

Model fit

Overall logistic regression model determines the extent to which the predicted values accurately represent the observed values (Xie, Pendergast, & Clarke, 2008)

- **Change in log likelihood**
- Hosmer and Lemeshow goodness-of-fit test
- Pseudo-variance explained
- **Predicted group membership**

Significance tests: Beta coefficients

- Each logistic regression coefficient determines if the individual coefficients are statistically significantly different from zero

Change in Log Likelihood test for model fit (1)

Test of overall model fit

- Useful when **comparing two <u>nested</u> models**
- The test is based on the change in the log likelihood function from the smaller model (with fewer variables) to a larger model (with the same variables in the smaller model and one or more additional variables)
- **Note**: this includes the intercept-only model with no predictors

The test is all regression coefficients are 0 → $H_0: \beta_1 = \beta_2 \ldots \beta_m = 0 \quad H_1: at\ least\ one\ \beta\ \neq 0$

The test statistic computed as -2 * $LL_{[diff]}$ is distributed as χ_2 with *df* = *df*[larger model] – *df*[smaller model]

The larger the difference, the better the model fit

Predicted Group Membership for determining model fit (2)

If the predicted probability is above a cutoff (usually .5) assign 1, otherwise 0

A crosstab table of predicted to observed probabilities provides the frequency and percentage of cases correctly classified

A perfect model produces 100% correctly classified cases

A model that classifies no better than chance would provide 50% correctly classified cases.

Terminology for Predicted Group Membership table (1)

Sensitivity is the probability that the dependent variable was coded as a “1” and our model predicted a “1” i.e., it was **classified correctly**

- The percentage of correct predictions of the cases that are coded as 1 for the dependent variable
- **Example**: You have the disease and we said you did

Specificity is the probability that a case coded dependent variable was coded as a “0” and our model predicted a “0” i.e., it was **classified correctly**

- The percentage of correct predictions of the cases that are coded as 0 for the dependent variable
- **Example**: you don’t have the disease and we said you didn’t

This should sound really familiar

Terminology for Predicted Group Membership table (2)

False positive rate is the probability that a case coded as 0 for the dependent variable (aka ‘negative’) is classified incorrectly.

- In other words, this is the percentage of cases in error where the dependent variable is predicted to be 1 (i.e., disease), but in fact the observed value is 0 (i.e., no disease)
- **Example**: you have the disease, but you don’t

False negative rate is the probability that a case coded as 1 for the dependent variable (aka ‘positive’) is classified incorrectly.

- In other words, this is the percentage of cases in error where the dependent variable is predicted to be 0 (i.e., unprepared), but in fact the observed value is 1 (i.e., prepared)
- **Example**: you don’t have the disease, but you we said you did

Test of Significance of the Logistic Regression Coefficients

The second test in logistic regression is the test of the statistical significance of each regression coefficient, b_k

$$H_0: \beta_k = 0 \qquad\qquad H_1: \beta_k \neq 0$$

The test statistic is called a Wald test computed as

$$W = \frac{\beta_k^2}{SE_{\beta_k^2}} \sim \chi_2$$

What Logistic Regression Is and How It Works: Assumptions

Noncollinearity

Linearity

Independence of errors

Values of *X* are fixed

Conditions

- Nonzero cell counts
- Nonseparation of data
- Lack of influential points
- Sufficient sample size

Recall the model

$$Y = \begin{cases} 0 & \text{If the condition is false} \\ 1 & \text{If the condition is true} \end{cases}$$

$$p(X) = P[Y = 1|X = x] = \frac{exp^{(\beta_0+\beta_1 X)}}{1 + exp^{(\beta_0+\beta_1 X)}}$$

Our goal is to estimate the unknown parameters, $\hat{\beta}_0$, $\hat{\beta}_1$

Review Algebra

Properties of exponents

$e^x * e^y = e^x + e^y$

$e^{x+y} = e^x + e^y$

Properties of logarithms

$\ln(e^x) = x$

$e^{\ln(x)} = x$

$e^0 = 1$

Mathematical Snapshot

$Logit(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_m X_m$	
$Logit(Y) = \beta_0 + \beta_1 X_1$	$\beta_0,\ \beta_1$ is interpreted as "log odds of Y"
$exp^{Logit(Y)} = exp^{\beta_0 + \beta_1 X_1}$	Take the exponent of each side
$Y = exp^{\beta_0 + \beta_1 X_1}$	This follows because $e^{\ln(x)} = x$ $\exp(\beta_0),\ \exp(\beta_1)$is interpreted as the "odds of Y"
$\Pr(Y) = \dfrac{exp^{\beta_0 + \beta_1 X_1}}{1 + exp^{\beta_0 + \beta_1 X_1}}$	This follows because $\dfrac{Odds}{1 + Odds}$ so that $\dfrac{Y}{1 + Y}$ is a probability
$\Pr(Y = 1 \mid X = x) = \dfrac{exp^{\beta_0 + \beta_1 X_1}}{1 + exp^{\beta_0 + \beta_1 X_1}}$	This is interpreted as a 'conditional' probability

Binary Logistic Regression in JASP/JAMOVI

Use the file gpa-college-enroll.JASP

Analyze → Regression → Binary Logistic

In this example we have

Y = college enrollment

X_1 = undergraduate GPA

$$logit(Y) = -2.144 + .833GPA$$

$$p(X) = Pr[Y = 1|X = x] = \frac{exp^{(-2.144 + .883X)}}{1 + exp^{(-2.144+.883X)}}$$

Probability of enrolling in college 'conditional on' values of one's grade point average

Coefficients

				Wald Test		
	Estimate	Standard Error	z	Wald Statistic	df	p
(Intercept)	-2.144	2.416	-0.888	0.788	1	0.375
gpa	0.883	0.821	1.076	1.157	1	0.282

Note. enroll level '1' coded as class 1.

The Wald test is the test statistic for the logistic regression model

The sig. is the *p*-value

EXP(B) is the odds ratio – the interpretation is: the odds of enrolling in college are about 2.5 times greater for each one unit increase in undergraduate gpa, but the result is not statistically significant

Computing probabilities

The most useful part of logistic regression is the ability to predict the conditional probability of Y given values of the independent variables

For example: what is the probability of college enrollment given that your gpa is 1.0, 2.0, 3.0 and 4.0?

$$\Pr(Y = 1) = p = \frac{e^{\beta_0 + \beta_1 X_1}}{1 + e^{\beta_0 + \beta_1 X_1}}$$

$$\Pr(Y = 1|X = 1) = \frac{e^{-2.144+.883(1)}}{1+e^{-2.144+.883(1)}} = \frac{e^{-2.144+.883}}{1+e^{-2.144+.883}} = \frac{e^{-1.261}}{1+e^{-1.261}} = \frac{.283371}{1+.283371} = .221$$

$$\Pr(Y = 1|X = 2) = \frac{e^{-2.144+.883(2)}}{1+e^{-2.144+.883(2)}} = \frac{e^{-2.144+1.766}}{1+e^{-2.144+1.766}} = \frac{e^{-.378}}{1+e^{-.378}} = \frac{.283371}{1+.283371} = .407$$

Table of predicted probabilities

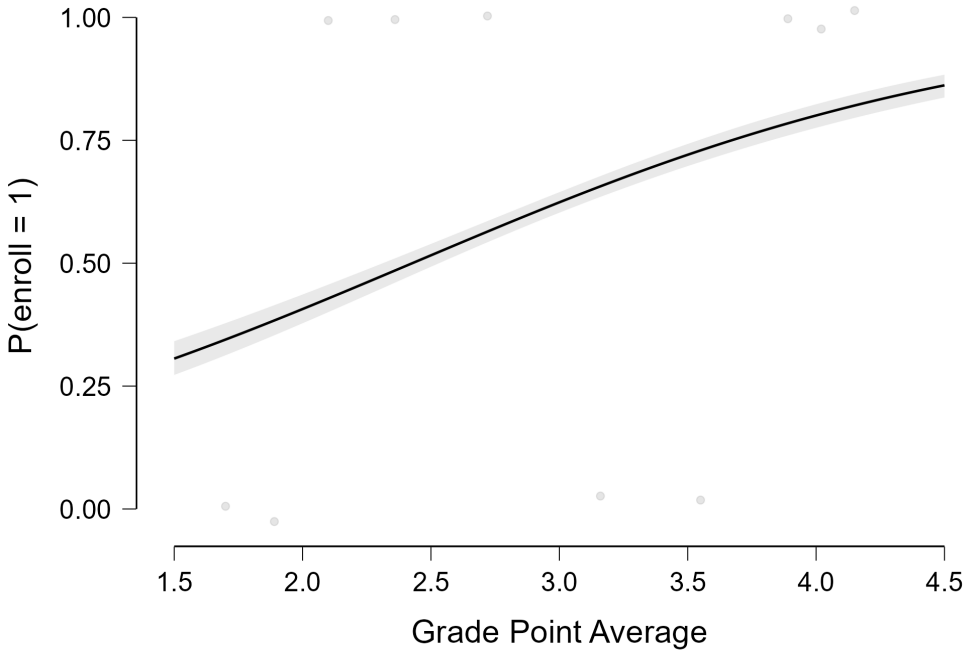
GPA (*X*)	β_0	β_1	$e^{\beta_0+\beta_1 X_1}$	$1+e^{\beta_0+\beta_1 X_1}$	$\frac{e^{\beta_0+\beta_1 X_1}}{1+e^{\beta_0+\beta_1 X_1}}$
1	-2.144	0.883	0.283370514	1.2833705	0.220801796
2	-2.144	0.883	0.685230501	1.6852305	0.406609363
3	-2.144	0.883	1.65698552	2.6569855	0.623633628
4	-2.144	0.883	4.006828377	5.0068284	0.800272763

See excel file gpa_college_enrollment_example.xlsx

Probability of enrolling in college conditional on high school gpa

As gpa increases the probability of enrolling increases

The probability is about .30 for a gpa = 1.5 but increases to about .9 for a gpa = 4.5



Your turn

- Open the file passing.JASP in JASP
- The data has three made-up variables:
  - Passed a statistics test
  - Hours of study
  - Gender (1 = male, 0 = female)
- Run a binary logistic regression of passed on hours
  - Write out the regression equation
  - Predict the probability of passing for 0, 50, 100, 150,200 and 250 hours of studying
  - Add gender into the model and interpret the coefficient in terms of odds and percent change in odds
  - Does adding gender improve the model fit?
  - Compute the probability of passing for males and females for 100 study hours

Model & Interpretation

$$logit(Y) = -9.3 + .082X$$

$$p(X) = P[Y = 1|X = x] = \frac{exp^{(-9.3+.082X)}}{1 + exp^{(-9.3+.082X)}}$$

Check: What is the Probability of passing when hours = *X*
Note: exp(.082) = 1.085

Table 1. Predicted probability of Passing Conditional on Hours Studying

X	0	50	100	150	200	250
P(*Y*|*x*=*X*)	0.0055	0.0603	0.4286	0.6298	0.8975	0.9783

Each additional hour spent working increases the odds of passing by [exp(.082)-1 *100] percent

Each additional hour spent working increases the odds of passing by about 8.5 percent

The odds of passing are 1.085 times higher for each additional hour spent studying

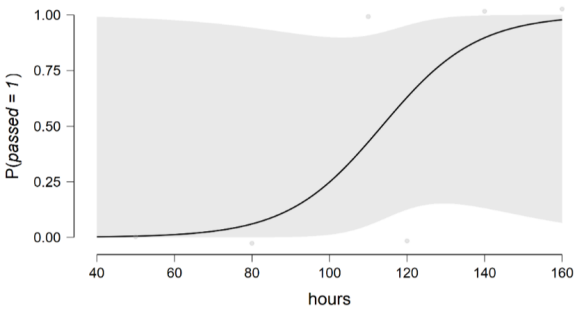
Model Summary - passed ▼

Model	Deviance	AIC	BIC	df	X^2	p	McFadden R^2	Nagelkerke R^2	Tjur R^2	Cox & Snell R^2
H_0	8.318	10.318	10.110	5						
H_1	4.078	8.078	7.661	4	4.240	0.039	0.510	0.676	0.536	0.507

Coefficients ▼

					Wald Test		
	Estimate	Standard Error	Odds Ratio	z	Wald Statistic	df	p
(Intercept)	−9.299	8.242	9.149e −5	−1.128	1.273	1	0.259
hours	0.082	0.070	1.085	1.169	1.367	1	0.242

Note. passed level '1' coded as class 1.



Jasp Output

Let’s add a binary variable, gender, to the equation

Coefficients

	Estimate	Standard Error	Odds Ratio	z	Wald Test		
					Wald Statistic	df	p
(Intercept)	−8.924	8.487	1.332e −4	−1.051	1.106	1	0.293
hours	0.079	0.071	1.083	1.126	1.267	1	0.260
gender (1)	−0.654	3.764	0.520	−0.174	0.030	1	0.862

Note. passed level '1' coded as class 1.

The odds of passing for males are [exp(-.654)-1 *100] percent lower compared to females

The odds of passing for males are 48.01% lower compared to females

The odds of passing are .5100 (exp(-.654)) times lower for males compared to females

The file probability of passing by gender.xlsx has the probability of passing for males and females given 100 hours of study time that I calculated by hand

Full example in R